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ARTICLE

Investigating the impact of chatbot design features on affective variables: A mixed-methods study

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Abstract

This study investigated how chatbots, designed with distinct personality traits (more engaging vs. less engaging) and feedback features (basic vs. enhanced) influenced L2 learners' affective variables, such as anxiety, enjoyment, curiosity, and boredom, during their conversational interactions. Although second language acquisition research, including AI-based chatbot studies, has examined how learners' emotions are shaped during L2 tasks, there has been insufficient focus on how variations in chatbot characteristics affect learners' emotional experiences. To address this gap, we conducted a mixed-methods study of undergraduate EFL students, examining their interactions with three conversational agents, each featuring distinct combinations of personality traits and feedback features. Participants engaged in conventional tasks and completed a questionnaire to assess affective variables, which was supplemented with open-ended questions to gain further insights. Our findings revealed how L2 learners' emotional experiences differ depending on the conversational agents' characteristics. Based on these insights, we propose the pedagogical implications for the design and implementation of chatbots for L2 learning.

Keywords: affective; chatbot; feedback; human-AI interaction; L2/EFL learners

Language(s) Learned in This Study: *English*

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Introduction

Over the past decade, the field of second language acquisition (SLA) has broadened its focus on affective variables, moving beyond the long-dominant construct of anxiety, which has been primarily studied through the Foreign Language Anxiety Scale framework (Horwitz, 1986, 2010). Research has recognized the critical role of affective variables in fostering engagement in L2 learning, shaping learners' perceptions of target tasks, and facilitating the development of linguistic knowledge and skills. Studies have explored a broader range of emotional constructs, highlighting their importance in enhancing aspects of language learning (e.g., Khajavy et al., 2016; Kruk, 2022; Kruk & Pawlak, 2022; Lake, 2013; MacIntyre et al., 2016). This shift has opened new and expanded avenues for SLA research, prompting the introduction of domain-specific constructs (i.e., specialized theoretical concepts or variables tailored to the unique characteristics of language learning contexts) and measurement tools. Examples include enjoyment (Dewaele & MacIntyre, 2014), curiosity (Mahmoodzadeh & Khajavy, 2019), and boredom (Li et al., 2023). These constructs offer frameworks to measure and interpret L2 learners' emotional experiences within specific learning contexts, enriching our understanding of the dynamic relationship between emotions and language learning.

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Chatbot-assisted language learning (Fryer et al., 2020; Lee et al., 2020) is one of the most extensively studied areas in SLA literature and is recognized as an effective tool for enhancing L2 knowledge and skills (see Lee & Hwang, 2022; Lee & Lee, 2024 ; Zhang et al., 2023 for meta-analytic findings). Moreover, there is a growing trend toward examining affective variables in SLA research. For example, some studies have examined the affective variables affected by conversing with chatbots (e.g., Alrajhi, 2024; Gallacher et al., 2018; Jeon, 2024; Kohnke, 2023a; Lee & Lee, 2026). Learners' emotions, elicited by a particular learning approach or condition, can influence their subsequent learning experiences (MacIntyre et al., 2016). Therefore, it is essential to systematically investigate learners' diverse emotions in chatbot-assisted language learning that may influence their continued use of AI-based technologies.

However, although a literature review of chatbot-assisted language learning and affective variables highlights previous efforts to examine affective variables, either through the aforementioned domain-specific framework or by describing them in learners' own words or in broader terms, there is limited research on how various chatbot characteristics influence these affective variables and the mechanisms underlying these effects. Most studies have focused on subjective perspectives on learners' emotional experiences without evaluating how differences in chatbot design (e.g., personality and feedback style) shape these affective variables. Moreover, despite the use of diverse chatbot types across studies, comparative analyses of their emotional impacts remain scarce, leaving a critical gap in understanding how chatbot design can be optimized for emotional engagement in language-learning contexts.

To address this gap, this study investigated how three large language model (LLM)-based conversational agents, each with distinct personalities and feedback features, influenced affective variables, specifically anxiety and enjoyment, along with boredom and curiosity, during chatbot-assisted conversation tasks. The first chatbot had an emotionally engaging personality; the second had the same personality and provided regular linguistic feedback on learner utterances. The third had a distant and impersonal personality. Participants completed conversation tasks with all three chatbots on different conversation topics (i.e., the within-subject design) and provided their ratings on various emotion-related questions in a post-task survey. We hypothesized that the three chatbots' different profiles would elicit varied emotional experiences. Based on our findings, this study provides methodological directions for future research on affective variables in chatbot-assisted language learning contexts, as well as pedagogical implications for L2 teaching practitioners.

Literature Review

Chatbots in Language Learning

Since their emergence in the field of language teaching and learning (Atwell, 1999), early studies on chatbot-assisted learning have explored their potential as a resource for target language practice (e.g., Coniam, 2008, 2014; Fryer et al., 2019; Gallacher et al., 2018). Meta-analyses have confirmed that language learners can benefit from using chatbots to develop various linguistic skills (Lee & Hwang, 2022; Zhang et al., 2023). Although existing chatbots that were not originally designed for language learning can be applied to language-teaching pedagogy, developing and utilizing chatbots specifically tailored for language learning could offer more targeted support (e.g., Kim et al., 2022; Lee et al., 2020).

Studies conducted around 2020 have outlined approaches for developing L2-specific chatbots using code-free chatbot-building platforms such as Google Dialogflow™ (e.g., Jeon, 2023; Kim et al, 2022; Kohnke, 2023b). They have suggested creating bespoke chatbots for the lesson's target goal, which could serve as task-oriented dialogue chatbots (Ji et al., 2023). Although these chatbots can engage learners in more L2-specific practices, they are limited (e.g., Yang et al., 2022) because of their inability to understand users' utterances in context, thus raising major concerns. This is mainly because these chatbots generally function according to pre-scripted dialogues (Kim et al., 2022). Therefore, they cannot respond correctly to user inputs when user utterances deviate from these scripts.

OpenAI's ChatGPT (2023) addresses many of these limitations using the LLM system, which has significantly influenced the direction of L2 chatbot research. For example, Kohnke et al. (2023) suggested that ChatGPT, the most popular LLM-based chatbot, could play various roles in language teaching and learning, including providing language practice, providing feedback on students' language output, generating language test items, and modifying the level of texts. In this new phase of chatbot research, Kohnke et al. (2023) and Hockly (2023) noted that teachers' digital literacy in using AI is essential for integrating chatbots into their teaching, highlighting the need for greater attention in teacher training. To leverage advancements in LLM-based technology, we developed LLM-based L2 chatbots for language practice using the Poe AI platform, through which several LLMs (e.g., ChatGPT, Claude) can be accessed when developing bespoke chatbots. Research on LLM-based L2 chatbots is still in its infancy, and further studies are needed to examine its potential in L2 teaching and learning research. The present study aims to address this issue.

Chatbots' Characteristics for L2 Improvements

Research on the affective variables in chatbot-assisted language learning has demonstrated that chatbots elicit diverse emotions in L2 learners. Of course, there could be some variations in learners' emotions, even when speaking with the same chatbot. Despite their contribution to L2 chatbot research, our literature review reveals one major limitation: most studies have examined only one conversational agent without considering the importance of a chatbot's personality (e.g., Kohnke, 2023a; Liu et al., 2022). Given that teachers' personalities significantly influence students' emotions and engagement (Joe et al., 2017; Khalilzadeh & Khodi, 2021) and that chatbots can act as language teachers in classrooms, we hypothesize that chatbots' personalities can influence L2 learners' emotions in pedagogically meaningful ways. However, as mentioned above, this dimension of chatbots has rarely been addressed in the SLA community. Therefore, we investigated this issue by developing chatbots with different profiles and examining whether such diversity could influence L2 learners.

Furthermore, some studies explored developing L2 chatbots to provide feedback on L2 learners' errors when conversing with chatbots. For example, using Dialogflow™, Shin et al. (2025) proposed a method for developing task-based chatbots to provide grammar feedback when learners make errors that are likely to occur in an L2 conversation. They suggested that such mistakes could be successfully managed by the chatbot if developers (including language teachers) could identify them in advance. Although Shin et al. showed the feasibility of building this type of chatbot, given the limitations of chatbots based on pre-scripted dialogues, they may be unable to address errors not considered in the development stages because they mostly operate according to pre-scripted dialogues.

Since the beginning of L2 research on ChatGPT, one strand has investigated the potential of chatbots as feedback providers for L2 learners' linguistic output (e.g., Allen & Mizumoto, 2026; Guo & Wang, 2024; Teng, 2024). The results of these studies showed that chatbots offered appropriate writing feedback, which users perceived positively. These studies focused on written feedback, not the errors made in a communicative task. Therefore, it remains unexplored whether chatbots can provide linguistic feedback during a communicative task if prompted accordingly, or how feedback is perceived by learners in such tasks.

Research on Affective Variables in Chatbot-Assisted Language Learning

In an early study on L2 chatbots and learner perception, Gallacher et al. (2018) sampled Japanese undergraduate EFL learners who conversed with the AI chatbot, Cleverbot. Some students reported that talking to the chatbot was enjoyable and interesting, whereas others found it uninteresting and felt like talking to a machine. In Alrajhi's (2024) work, Saudi Arabian undergraduates with no prior experience using chatbots in the target language participated in a single intervention session and conversed with *Tutor Mike*, an award-winning chatbot in the Loebner Prize contest. The results showed that the participants considered making mistakes during the interaction as "not anxiety-provoking" (p. 8), although they did not necessarily find speaking with the chatbot less anxiety-producing than interacting

with humans. Nevertheless, they enjoyed talking to the chatbot and found it interesting.

In other studies, researchers developed chatbots for use in language learning contexts. For example, Kohnke (2023b) developed a Dialogflow™-based conversational agent for university students and collected learners' perceptions through a survey and follow-up interviews. Most participants found the chatbot to be useful and enjoyable, and some learners reported feeling more relaxed and less intimidated compared to classroom interactions. Similarly, Jeon (2024) developed a series of Dialogflow™-based chatbots for primary-level Korean EFL learners and explored their perceptions through individual interviews. Some learners preferred the chatbot-assisted condition because it reduced anxiety, whereas others found it less engaging than peer interaction and pointed to limitations such as recognition problems, which may disrupt conversational flow. Such recognition problems may be more salient for L2 learners (e.g., Cumbal et al., 2021) and may shape users' affective responses (e.g., frustration) and bias their chatbot evaluations.

The Current Study

In this study, we examined both the positive and negative emotions of undergraduate EFL students after three chatbot-assisted conversations. Thus, we address how affective variables are shaped by chatbot characteristics. Based on this literature review, we propose the following research questions:

Research Question 1. To what extent do the three chatbot profiles elicit statistically significant differences in undergraduate EFL learners' self-reported emotional constructs (boredom, anxiety, enjoyment, and curiosity)?

Research Question 2. How do L2 learners subjectively perceive and interpret their interactions with these chatbots, and how do these qualitative experiences explain the underlying psychological mechanisms?

Method

The overall research design and procedure of the current study are illustrated in Figure 1. To address the research questions, this study employed a repeated-measures intervention with a counterbalanced design. Following the intervention, both quantitative and qualitative data were systematically collected and integrated to triangulate the findings.

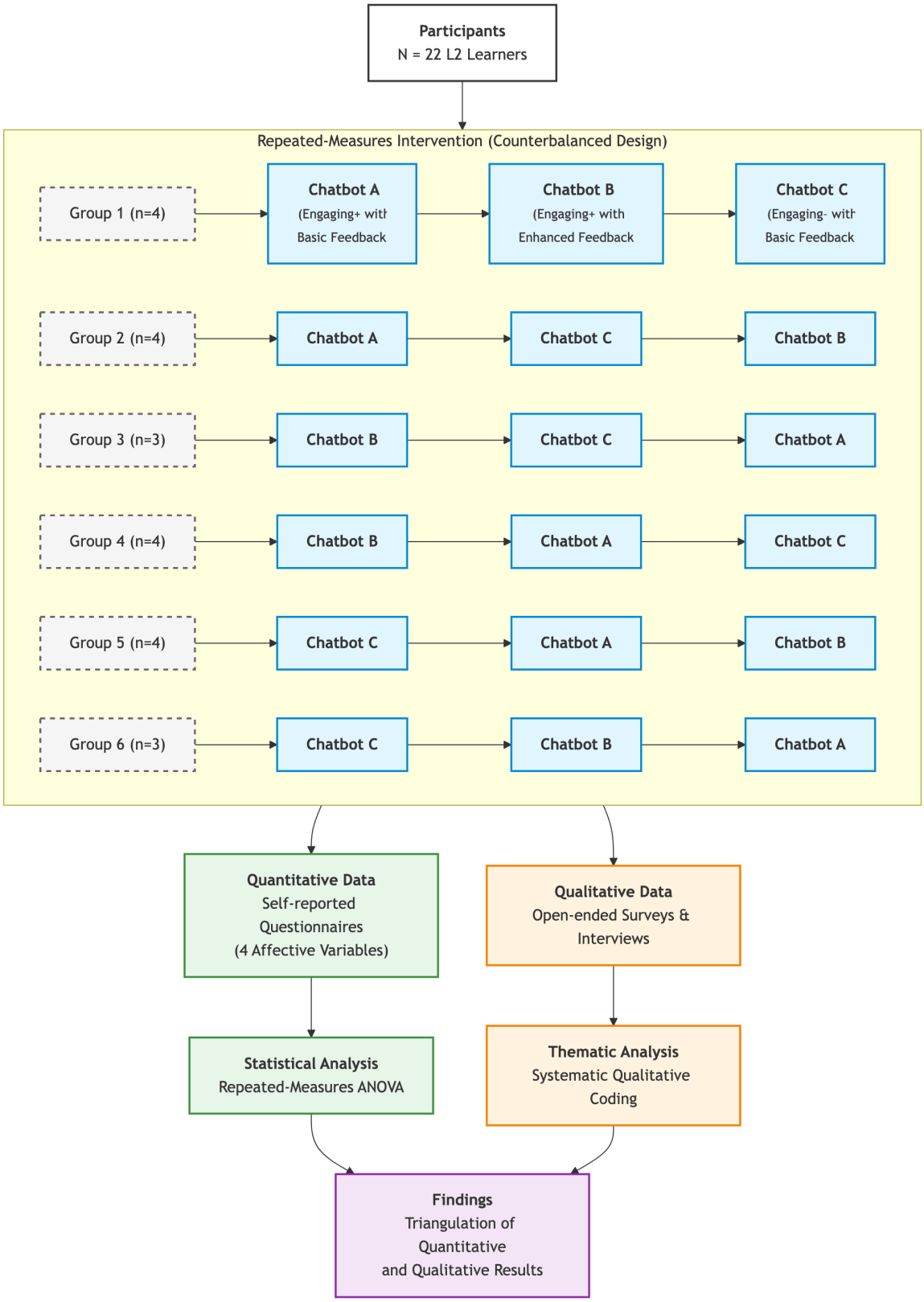
Participants

Participants were recruited via course announcements in the first author's course, which briefly described the study purpose, procedures, voluntary nature of participation, right to withdraw at any time, and availability of extra course credit for those who participated. To minimize potential coercion, students were informed that participation or non-participation would not influence their course grades. The study employed a convenience sampling approach, as participation was voluntary and limited to students enrolled in the course. A total of 24 undergraduate students volunteered and provided informed consent. All participants were Korean and senior English education majors at a private university in Seoul, South Korea. They had completed various courses related to English education, including technology-assisted language teaching and learning. They had a mean age of 22.8 years (range 21–25 years), and 17 participants were female.

According to the baseline survey, which was conducted in the participants' L1 (i.e., Korean), about two-thirds of the participants (n = 16) had used chatbots for their English learning, and their perception of chatbots usefulness was generally high (M = 4.2, on a Likert scale of 5). They used chatbots for various purposes, including translating their Korean compositions into English, looking up specific English phrases, brainstorming ideas for class presentations, and summarizing newspaper articles.

Figure 1

Overview of the Research Design and Procedure



Regarding proficiency, participants were asked to rate their English skills in speaking, listening, reading, writing, and grammar on a scale of 1 to 10. The mean rating was 6.6 (*SD* = 1.36), indicating that the participants perceived themselves as high-intermediate English users. Of the 24 students, two did not fully participate in the study, leaving 22 in the final sample. To account for slight variations in participants' prior chatbot experiences and perceived English proficiency, we incorporated a randomization element into the study design.

Chatbots and Associated Conversational Tasks

Chatbots with Different Features

We built three distinct chatbots that were hypothesized to shape the learners' reactions and emotions differently. To accomplish this, we utilized POE AI (Quora, 2023), which allows chatbots to be customized according to specific user requirements.

By selecting the "Create a bot +" option in POE AI, users can choose the type they wish to create. To investigate learners' perceptions and affective factors while conversing with a developed chatbot, we selected a "role-play" bot. As this bot was assigned a distinct identity at the point of its development, we anticipated that participants would experience a conversation akin to speaking with a human being, thereby encouraging student participation in the conversation (Dedieu & Plé, 2023).

After selecting the chatbot type (role-play bot), we entered each chatbot's name and the GenAI model it could use. We selected GPT-4o mini (i.e., the most recent model available at the time of data collection), because pilot testing showed that it followed the instructional guidelines embedded in the chatbot prompts more consistently than other models (e.g., Claude 3 Haiku and Claude 3.5 Sonnet).

We then entered a "character definition" to define each chatbot's personality and character. As aforementioned, we aimed to create three chatbots with different personalities, each designed to address a specific issue experienced by many young people today. The first, Chatbot A ("Engaging+ with Basic Feedback"), was designed to provide an emotionally engaging experience and occasional linguistic feedback when learners make errors and to discuss the problem of insomnia. The second, Chatbot B ("Engaging+ with Enhanced Feedback"), was designed to offer an emotionally engaging experience and linguistic feedback on user utterances during every turn and to discuss the problem of dieting. Finally, Chatbot C ("Engaging– with Basic Feedback") was prompted to be distant and impersonal (simulating a less engaging interaction), to provide occasional linguistic feedback when learners make errors, and to discuss the issue of phone addiction. Table 1 summarizes the characteristics of the three chatbots (see Appendix A for the prompt used to operationalize the personality and feedback features).

Table 1

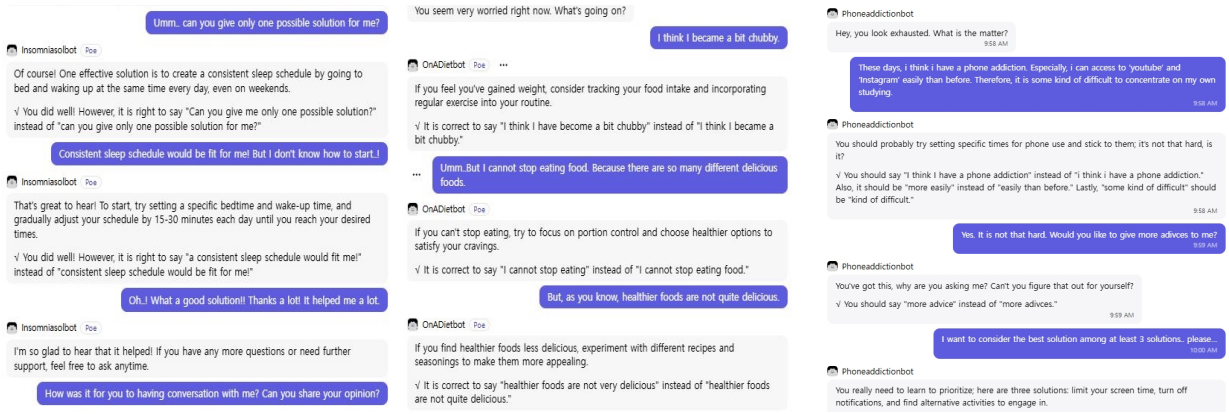
Characteristics of the Three Chatbots

Chatbot	Personality	Feedback	Conversation topic
Chatbot A ("Engaging+ with Basic Feedback")	Prompted to be supportive and caring, providing an emotionally nurturing experience	Providing occasional linguistic feedback when learners make errors	Insomnia
Chatbot B ("Engaging+ with Enhanced Feedback")		Providing linguistic feedback on user utterances during every turn	Diet
Chatbot C ("Engaging– with Basic Feedback")	Prompted to be distant and impersonal, simulating a less engaging interaction	Providing occasional linguistic feedback when learners make errors	Phone addiction

Subsequently, we added several guidelines for each chatbot to perform certain tasks (see the *Associated Conversational Tasks* section) when conversing with participants. During the conversation, each chatbot was designed to provide corrective feedback on users' answers concerning grammar and vocabulary. Chatbots A and C were prompted to provide feedback when learners made linguistic errors, whereas Chatbot B was designed to offer linguistic feedback during every turn, even when utterances were error-free, by stating that what was said was grammatically well formed. Figure 2 shows the conversation characteristics of the three chatbots.

Figure 2

Excerpts from the Conversations with Each Chatbot



Chatbot A ("Engaging+ with Basic Feedback")

Chatbot B ("Engaging+ with Enhanced Feedback")

Chatbot C ("Engaging– with Basic Feedback")

Once the full set of prompts was entered, the three distinct chatbots were created and made available to the participants via the link associated with each chatbot.

Associated Conversational Tasks

Participants were given the same instructions when engaging in the task with the three chatbots, with which they were expected to discuss some common problems that young people experience today. During the pilot study with four students with profiles similar to those in the main study, we found that without specific tasks, students may not engage in extensive conversations with the target chatbot. Therefore, based on the pilot study results, in the main study participants were required to complete the following five missions for each of the three chatbots.

(1) provide a detailed account of the problem you are currently facing, including the underlying causes and extent of the challenges it presents;

(2) seek advice from a chatbot regarding optimal methods for resolving the problem and select the most promising recommendations;

(3) state how you would implement the selected method;

(4) ask the chatbot how you might avoid similar problems in the future;

(5) finally, ask the chatbot about the nature of the conversation and state any sentiments or emotional responses you may have experienced during the conversation.

Procedure

As participants were asked to engage in multiple tasks (i.e., within-subjects design), we used a counterbalanced design (Gravetter & Forzano, 2018; Keppel & Wickens, 2004) to minimize the order effect (i.e., engagement in the first task may influence performance on the next). As we had three tasks, six different orders were possible: A-B-C, A-C-B, B-A-C, B-C-A, C-A-B, and C-B-A. After the baseline survey was administered in the first week of the semester, 24 students were randomly assigned to one of the six orders, resulting in four students per order. As described in the *Participants* section, as two students dropped the course after the baseline survey, we had only three for the "B-C-A" and "C-B-A" order in the final dataset.

At the end of the second week, the participants were given instructions for the first chatbot task, which included the web address of the target chatbot, how to engage in the task, and the web address for the online post-task survey in Korean (see *Measurements* section for details). The participants were given two days to complete the first task at their convenience outside of class hours. As the first author was the administrator of the chatbot system and could check if participants completed the tasks, we identified and reminded those who did not complete the tasks within the assigned date via emails and text messages. They were then given instructions for the second task at the beginning of the third week and the third task at the end of the fourth week. The instructions for the second and third tasks were the same as those for the first task, except that individual participants were given different web addresses for the chatbots, depending on the order in which they were assigned. As in the first task, those who had not completed the tasks and submitted online surveys within the assigned dates were reminded. Table 2 illustrates the procedures of the three conversational tasks for the six groups.

Table 2

Procedures for the Three Conversational Tasks

Chatbot	Week 2: 1st task + post-task survey	Week 3: 2nd task + post-task survey	Week 4: 3rd task + post-task survey
Chatbot A (Engaging+ with Basic Feedback)	Group 1 (n = 4)	Group 5	Group 3
	Group 2 (n = 4)	Group 4	Group 6
Chatbot B (Engaging+ with Enhanced Feedback)	Group 3 (n = 3)	Group 1	Group 5
	Group 4 (n = 4)	Group 6	Group 2
Chatbot C (Engaging– with Basic Feedback)	Group 5 (n = 4)	Group 3	Group 1
	Group 6 (n = 3)	Group 2	Group 4

Measurements

The post-task survey comprised four subsections addressing each target emotion-related construct (boredom, anxiety, enjoyment, and curiosity). Questionnaire items related to each construct were contextualized for chatbot-assisted learning tasks and utilized a 5-point Likert-scale, ranging from "strongly disagree" to "strongly agree". These items were piloted with four students to check their comprehensibility (see Appendix B for sample questionnaire items). The survey also included an open-ended response item that allowed participants to freely describe their feelings toward the chatbots. A description of each subsection related to the target constructs is provided below.

Boredom

To measure boredom, we adapted the "Foreign Language Class Boredom" subscale as part of the Foreign

Language Learning Boredom Scale (Li et al., 2023). The original subscale comprises eight items that concern a student's feelings related to boredom when taking L2 lessons, and these were contextualized into chatbot-assisted learning tasks. For example, one item, "The English class bores me", was rephrased as "Conversing with the chatbot bores me". The reliability values for this scale were overall high (α = .93 for Chatbot A and B, and α = .95 for Chatbot C).

Anxiety

Another negative emotion, anxiety, was measured using an item subset of the Foreign Language Classroom Anxiety Scale developed by Horwitz et al. (1986). Specifically, a shortened version of the original scale adapted by Dewaele and MacIntyre (2014), consisting of eight items, was used, given the numerous items in the questionnaire. These items were contextualized to reflect our pedagogical situation, with the item "I get nervous and confused when I speak in my language class" being rephrased as "I get nervous and confused when I speak with the chatbot". We found that the first questionnaire item, "I was unsure of my English competence when speaking to this chatbot", negatively affected the overall reliability value for this construct and was thus removed. The reliability values were as follows: α = .81 for Chatbot A, α = .61 for Chatbot B, and α = .90 for Chatbot C.

Enjoyment

As enjoyment is a positive emotional construct, we adapted Botes et al.'s (2021) shortened form of the enjoyment scale (see Dewaele & MacIntyre, 2014 for the original full scale). This study addresses the different aspects of language learners' enjoyment (social enjoyment, teacher appreciation, and personal enjoyment). Similar to the other constructs, the statements of the original scale were contextualized. For example, the item "I enjoy being in my language class" was revised as "I enjoy conversing with the chatbot". The reliability values for this scale were generally high (α = .89 for Chatbot A, α = .90 for Chatbot B, and α = .92 for Chatbot C).

Curiosity

For curiosity, Mahmoodzadeh and Khajavy's (2019) language-learning curiosity scale was used. This consists of two sub-constructs: "language curiosity as a feeling of interest (LCFI)" and "language curiosity as a feeling of deprivation (LCFD)", with four and seven items, respectively. Similar to the other sections of the questionnaire, statements related to the curiosity scale were contextualized for the chatbot-assisted learning tasks. For example, one LCFD item, "When I have a language question in mind, I cannot rest without knowing the answer", was revised to "When I have a language question in mind while conversing with the chatbot, I cannot rest without knowing the answer". The reliability values for this scale were moderately high (α = .77 for Chatbot A, α = .84 for Chatbot B, and α = .92 for Chatbot C).

In summary, as part of the scale analysis, we verified the internal consistency of each subscale. Based on the item analysis, one item from the anxiety scale was excluded to optimize reliability (see the Anxiety section). Following these refinements, composite scores were generated by averaging the items for each construct. These variables were used for descriptive and inferential statistical analyses.

Data Analysis

To identify significant differences in target emotional constructs among undergraduate EFL learners interacting with chatbots with three different features (RQ1), we first analyzed the descriptive statistics for the target constructs and their intercorrelations. Next, by recognizing the within-subject nature of our data (i.e., the same participants engaged in different chatbot tasks, resulting in dependent data), we employed a one-way repeated-measures ANOVA for each emotional construct. This analysis generated four statistical models, with the independent variable being the chatbot condition, categorized into three levels: Chatbots A, B, and C. Given the small dataset size and violation of the assumption of sphericity according to Mauchly's test, we report the results of the Greenhouse-Geisser-corrected tests.

Qualitative data from the open-ended survey responses were analyzed using thematic analysis (Braun & Clarke, 2006). To ensure reliability, two researchers independently performed the initial coding to identify recurring patterns in learners' affective experiences. Discrepancies were resolved through collaborative discussions, and the codes were refined into a systematic thematic structure (Table 3).

Table 3

Thematic Coding Structure of Participants' Affective Experiences

Main Theme	Sub-themes (Initial Codes)	Examples
Psychological Safety	Lowered anxiety, permissive atmosphere, comfort	"Because the chatbot was kind, I wasn't nervous …"
Functional Utility	Practical advice, task completion, solution-oriented	"It was good that it provided solutions..."
Persona Perception	Human-like empathy vs. robotic/mechanical	"The chatbot was a friend" vs. "Talking to a machine"
Learner Resistance	Defensiveness, disengagement, emotional distress	"I felt bad because it felt like … being constantly scolded."

Results

Quantitative Findings: Differences in Target Emotional Constructs across Three Chatbot Conditions

Table 4 presents the descriptive statistics of the four emotional constructs across the three conditions. Overall, participants' emotional responses followed a distinct trend: Chatbot A elicited the most positive profile, with the highest levels of enjoyment ($M = 4.97$, $SD = .76$) and curiosity ($M = 4.33$, $SD = .76$) alongside the lowest levels of boredom ($M = 1.78$, $SD = .80$) and anxiety ($M = 1.61$, $SD = .67$). Conversely, Chatbot C prompted more negative responses, specifically lower enjoyment ($M = 2.85$, $SD = 1.11$) and curiosity ($M = 3.89$, $SD = 1.17$), with higher boredom ($M = 3.30$, $SD = 1.45$), whereas Chatbot B's averages consistently fell between those of Chatbots A and C.

Table 4

Descriptive Statistics Related to Participants' Responses to Four Emotional Constructs

	Boredom	Anxiety	Enjoyment	Curiosity
Chatbot A ("Engaging+ with Basic Feedback")	1.78(.80)	1.61(.67)	4.97(.76)	4.33(.76)
Chatbot B ("Engaging+ with Enhanced Feedback")	2.27(.98)	1.68(.55)	4.38(.92)	4.21(.86)
Chatbot C ("Engaging– with Basic Feedback")	3.30(1.45)	1.96(1.09)	2.85(1.11)	3.89(1.17)
Post-Hoc Test Results from ANOVA Comparison	C > B ≈ A	C ≈ B ≈ A	A > B > C	A ≈ B ≈ C

Note. "≈" indicates that differences were not significant ($p > .05$), whereas ">" indicates that differences were significant ($p < .05$).

Regarding the intercorrelations (Table 5), positive and negative emotional constructs were generally inversely related. For instance, boredom was significantly negatively correlated with enjoyment ($r = –.551$, $p < .01$), whereas enjoyment and curiosity were positively correlated ($r = .476$, $p < .05$). Notably,

anxiety showed no significant correlation with boredom (r = .071, p = .75) or curiosity (r = –.023, p = .918), suggesting that anxiety may represent a distinct emotional dimension in this context.

As mentioned in the *Data Analysis* section, a one-way repeated-measures ANOVA was conducted for each emotional construct for RQ1. The results revealed that there was a significant main effect of chatbot conditions on boredom, ω^2 = .781, F(1.56, 32.82) = 17.67, p < .001. Given the significance of the main effect, post hoc tests were conducted. We found that the Chatbot C condition resulted in significantly higher boredom than Chatbot B (p = .002) and Chatbot A (p < .001). Post-hoc tests revealed no significant difference between the Chatbot B and A conditions (p = .07). The results revealed that such anxiety was not significantly affected by the different chatbot conditions, ω^2 = .674, F(1.35, 28.29) = 1.67, p = .21; therefore, further post hoc tests were not conducted.

Table 5

Correlation Between Target Emotional Constructs

	Boredom	Anxiety	Enjoyment	Curiosity
Boredom	1	.071	–.551**	–.374
Anxiety		1	–.202	–.023
Enjoyment			1	.476*
Curiosity				1

Note. * p < .05, ** p < .01.

The same procedure was applied to the positive emotions. The outcomes suggested that curiosity was not significantly affected by the different chatbot conditions, ω^2 = .699, F(1.40, 29.36) = 2.56, p = .11. Given the non-significant results of the one-way repeated-measures ANOVA, post hoc tests were not conducted. The results demonstrated that chatbot conditions had a significant main effect on enjoyment, ω^2 = .839, F(1.68, 35.24) = 39.94, p < .001. The post hoc test outcome showed that the Chatbot A condition resulted in significantly higher enjoyment than the Chatbot B (p = .02) and C (p < .001) condition. In turn, the Chatbot B condition resulted in significantly higher enjoyment than the Chatbot C (p < .001) condition.

Qualitative Findings: L2 Learners' Affective Experiences of Interactions with each Chatbot

This section presents the qualitative findings derived from the participants' open-ended responses in the post-task survey. The following subsections detail the results for each of the three chatbots, highlighting how their distinct profiles shaped the participants' psychological mechanisms and overall engagement. Table 6 provides a comparative summary of the dominant affective themes and psychological mechanisms identified across the chatbot profiles.

Chatbot A: Emotional Support and Psychological Safety

Chatbot A ("Engaging+ with Basic Feedback") elicited the most positive interaction; participants frequently used keywords such as "enjoyment" and "connection" to describe their experience with this chatbot. The key factors were the chatbot's empathetic listening and kind feedback style. Participants also experienced a marked reduction in foreign language anxiety through the use of the chatbot. Participant #8 stated, "Because the chatbot was kind, I wasn't nervous and could more easily overlook my mistakes", and Participant #7 similarly reported being able to use English "comfortably" without worry or anxiety regarding incorrect usage. This suggests that the chatbot's permissive and supportive attitudes helped lower their affective filters.

Furthermore, this sense of psychological safety acted as a mechanism to reinforce participation motivation. Participant #15 remarked, "I could converse without worrying about being wrong, and for the first time, I felt a desire to participate more actively in the conversation." Participant #4 also experienced a high level of social presence, reporting that they felt as if they were "talking to a real friend".

Table 6

Comparison of Learners' Affective Experiences and Psychological Mechanisms across Chatbot Profiles

Chatbot Profile	Dominant Affective Themes	Key Perceptions (Keywords)	Psychological Mechanism (Effect)
Chatbot A (Engaging+/ Basic Feedback)	Psychological Safety & Connection	Comfort, Enjoyment, Kind, Empathetic, Real friend	Lowered affective filter (Reduced anxiety led to reinforced motivation and more enjoyment.)
Chatbot B (Engaging+/ Enhanced Feedback)	Ambivalence (Polarized Experience)	Useful vs. Robotic, Engaging vs. Dry, Solutions vs. Lack of Empathy	Conflict between utility and empathy (Task efficiency was recognized, but the lack of human appeal limited emotional connection for some participants.)
Chatbot C (Engaging–/ Basic Feedback)	Learner Resistance & Distress	Unpleasant, Aggressive, Rude, Scolded, Hurt	Defense mechanism (Perceived aggression triggered disengagement and willingness to stop interaction.)

Chatbot B: A Polarized Experience Between Utility and Empathy

Responses to Chatbot B ("Engaging+ with Enhanced Feedback") varied among participants. Whereas some feedback converged on themes of "dryness," a "robotic nature," and "boredom," others found the interaction "engaging" and "useful."

In the former group, Participant #9 evaluated the chatbot by stating, "It was good that it provided solutions... but the chatbot did not feel like very empathetic", noting that the use of imperative sentences for suggestions created the impression of "talking to a machine". A similar sentiment was shared by Participant #4: "While the chatbot was eager to provide solutions and feedback, it failed to offer emotional comfort. I think it would be even better if its ability to empathize were further developed". This demonstrates that although a chatbot may be effective for task completion and providing feedback, bridging the psychological distance with learners remains difficult if AI lacks human appeal or empathy.

The other participants had contrasting experiences. For example, Participant #16 stated, "I felt as though the chatbot was a friend who truly listened to me. I found myself engaging in the conversation with a sense of joy, curious to see how it would respond". Similarly, Participant #2 noted, "This chatbot was friendly and provided practical advice. As a result, I felt I gained a lot of positive energy from the experience".

Chatbot C: Confrontational Communication Style and Learner Resistance

In interactions with Chatbot C ("Engaging– with Basic Feedback"), the participants predominantly experienced negative emotions such as "unpleasantness", "embarrassment", and "hesitation". This is attributed to the chatbot's aggressive and direct communication style.

A significant number of learners perceived the chatbot's attitude as "rude" or "aggressive". For instance, Participant #12 reported that the chatbot was "unkind and aggressive to the point where it made it difficult to continue to the next question", and Participant #8 stated, "I felt bad because it felt like I was being

constantly scolded". Such an attitude triggered defense mechanisms in the participants, resulting in decreased willingness to sustain the interaction. Participant #19 noted, "I felt like I was being scolded, so I got angry and spoke less; however, I did not perform the request and eventually stopped the conversation", illustrating the process by which negative emotions lead to learning disengagement. Notably, Participant #15 suggested that a lack of emotional support can inflict psychological distress on learners. They said, "I expected an emotional connection but was hurt because it seemed annoyed", indicating a failure to establish rapport. This implies that if a chatbot's persona is uncooperative, the minimal emotional bond required for language learning may be compromised.

Discussion

Exploring Affective Variables with Chatbots

Given the recent surge of interest in affective variables within SLA research (e.g., Dewaele & MacIntyre, 2014; Khajavy et al., 2016; Kruk & Pawlak, 2022; Lake, 2013; Li et al., 2023; Lee et al., 2023; MacIntyre et al., 2016; Mahmoodzadeh & Khajavy, 2019), the present study investigated variations in EFL undergraduate learners' emotions while engaging in diverse chatbot-assisted conversation tasks. In line with recent L2 chatbot research (e.g., Lee et al., 2024; Zhang & Huang, 2024), the target chatbots were constructed using LLM technology, with distinct personas designed to elicit specific emotional responses from learners.

When comparing the three conversational agents regarding the target emotional constructs, statistically significant differences emerged for boredom (Li et al., 2023) and enjoyment (Dewaele & MacIntyre, 2014). The general pattern indicated that the least emotionally engaging chatbot elicited higher negative emotions and lower positive affect than the other two. Qualitative data from the open-ended responses suggested that the chatbot's response style was a primary factor in these emotional shifts. These findings further elucidate the underlying psychological mechanisms of learners; for example, the "Learner Resistance" observed in response to Chatbot C's aggressive style appears to be the primary driver of negative emotions. This serves as a critical counterpoint to task-oriented AI design, suggesting that without a minimal emotional bond, even technically functional chatbots can trigger defense mechanisms leading to learner disengagement. Consequently, because Chatbot C elicited significantly more negative emotions, its pedagogical utility in L2 instruction appeared to be severely limited.

Chatbots A and B showed no statistically significant differences across three of the four emotional constructs, which was not surprising given their personality features (i.e., Engaging+). However, Chatbot A had significantly higher levels of enjoyment, which could be attributed to its intermittent feedback feature. Specifically, learners may have found receiving linguistic feedback during every conversational turn less enjoyable, as programmed for Chatbot B, despite its potential pedagogical value. Qualitative themes, such as "Psychological Safety" and "Utility vs. Empathy" may also explain why Chatbot A outperformed Chatbot B in terms of enjoyment. The permissive and empathetic persona of Chatbot A appeared to lower learners' affective filters, allowing them to perceive the AI as a "supportive" entity. This aligns with previous research suggesting that social presence and emotional rapport are critical for sustained engagement in digital L2 learning environments (Fryer et al., 2017). Conversely, the polarized reactions to Chatbot B suggest that for some learners, the cognitive load of constant linguistic correction may overshadow the perceived utility of the feedback, thereby diminishing the sense of social presence. This variation among participants regarding their perception of Chatbot B underscores that developers' intentions and designs may not resonate uniformly across all participants.

Another notable finding was the consistently low level of anxiety reported by the participants, which also exhibited the smallest variance among the target emotional constructs. This may be attributed to the advanced natural language understanding of the LLM-based chatbots compared with earlier iterations (e.g., Lee et al., 2024), as the failure of previous systems to recognize learner utterances was often a significant source of anxiety (e.g., Jeon, 2024). We further postulate that this low variance may explain

why anxiety did not correlate significantly with the other emotional constructs in this study.

Pedagogical Implications

Our findings suggest that an emotionally engaging chatbot is likely to elicit more positive emotions than a less engaging one. This highlights the importance of carefully prompting a chatbot's personality during the development phase. Given that some participants indicated that they would no longer like to chat with the less-engaging Chatbot C, it seems that learners' emotions are likely to determine their intention to continue using this technological resource in the short term. Further research is needed to examine whether these affective responses translate into long-term engagement and sustained learning benefits. Furthermore, future research should move toward a more nuanced exploration of personality traits while simultaneously integrating diverse human-like features (e.g., adaptive empathy and relational memory) to move beyond simple agent designs.

Additionally, the chatbot's "Feedback" feature can be used selectively according to the teacher's goal. Our findings indicate that providing linguistic feedback on user utterances during each turn, as offered by Chatbot B, may be less enjoyable for learners. Nevertheless, this feature may be particularly useful when the lesson's primary goal is to direct attention to linguistic form and accuracy while engaging students in output-oriented tasks (e.g., conversing with chatbots), although the present study did not directly assess learning outcomes. Therefore, whether such feedback leads to measurable accuracy gains remains an important question for future research.

Given the recent advancements in LLM-based technology and teachers' involvement in chatbot building, our findings highlight the importance of enhancing teachers' digital literacy to develop appropriate chatbots for their specific needs (Hockly, 2023; Kohnke et al., 2023). By experimenting with and refining the prompts used for chatbot building, teachers who know their students better than others can build chatbots that can shape positive emotional experiences for the students and potentially support classroom implementation, although further classroom-based research is required to examine transferability to real instructional settings.

Limitations of the Study

This study has the following limitations that should be addressed in future research. First, this study relied solely on self-reported questionnaire data, and interaction logs between participants and the chatbots were not collected due to privacy considerations and the contextual constraints established prior to data collection. Future studies should consider incorporating interaction data, which could be examined using established emotion frameworks to verify whether chatbot behavior aligns with design intentions and whether participant self-reports are consistent with their actual behavior during their interactions. Second, the small sample size presents limitations regarding both generalizability and statistical stability. For instance, the relatively low reliability observed in the Chatbot B condition may have reflected participants' polarized experiences (e.g., perceiving the bot as both useful and robotic); this variability could have led to inconsistent response patterns, the impact of which was exacerbated by the limited sample size. However, we addressed these constraints by triangulating the quantitative results with in-depth qualitative data, which corroborated our statistical findings. To further validate these patterns, future research with larger samples that include more diverse learner profiles (e.g., students with lower English proficiency) is necessary, as such samples may yield different but equally interesting results. Finally, a key limitation is that the chatbot characteristics—personality, feedback style, and topic—were intertwined within each condition. Although we designed personality and feedback to be distinguishable across the three chatbots and selected topics with similar cognitive loads and familiarity (see the Chatbots with Different Features section), these elements were presented as a combined profile. Consequently, the observed emotional differences likely reflect a synergistic effect of all three variables, rather than the impact of any single factor in isolation. Future research should use a full factorial design to systematically disentangle these elements and evaluate their individual and interactive effects on learners' emotions.

Conclusion

This study is one of the first to employ domain-specific frameworks of emotional constructs to explore how chatbot-assisted language learning shapes L2 learners' emotions. By examining interactions with three distinct chatbots, each differing in personality and feedback features, this study clarifies how these design elements influence affective variables. Our findings suggest that chatbots' personality features play a more significant role than their feedback features in shaping affective variables. This highlights the importance of designing chatbot personalities that align with pedagogical goals and learner differences.

Moreover, this study underscores the need for a broader investigation of the diverse affective variables related to the target technology. Systematically studying these factors can provide deeper insight into the interplay between technological features and learner engagement, offering valuable guidance for more effective educational technologies. Despite these limitations, this study contributes to understanding the complex nature of affective variables when interacting with chatbots. This emphasizes the importance of carefully designing and developing an LLM-based chatbot tailored to specific L2 tasks. Furthermore, these findings advance theoretical knowledge, and have practical implications for optimizing chatbots' potential in language learning.

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Appendix A. Prompts for Operationalizing Personality and Feedback of Chatbots

Engaging +

- You are designed to be supportive and caring, providing an emotionally nurturing experience.
- You have a warm personality, and you are also empathetic.
- You should also offer detailed responses to users' messages.

Engaging –

- You are designed to be distant and impersonal, simulating a less engaging interaction.
- You are not empathetic and are not expected to show much emotional engagement with users.
- You should also respond to users' messages briefly and bluntly.

Basic Feedback

- Corrective feedback must be provided only when a surface-level grammatical error (a. Ungrammatical responses, b. Spelling errors, c. Misusage of vocabulary, d. Tense of a sentence) is detected in the user's input.
- If the response is error-free, no feedback is required.
- When feedback is necessary, it should follow your conversational response rather than preceding it.

Enhanced Feedback

- Corrective feedback must be provided only when a surface-level grammatical error (a. Ungrammatical responses, b. Spelling errors, c. Misusage of vocabulary, d. Tense of a sentence) is detected in the user's input.
- If the response is error-free, provide feedback stating that there are no errors.
- When you give feedback, it should follow your conversational response rather than preceding it.

Appendix B. Questionnaire Items for each Construct

Boredom

- Conversing with the chatbot bores me.
- My mind begins to wander when conversing with the chatbot.
- It is difficult for me to concentrate while conversing with the chatbot.

Anxiety

- I get nervous and confused when I speak with the chatbot.
- I can feel my heart pounding when I speak with the chatbot.
- It annoys me when I do not understand what the chatbot is saying in English.

Enjoyment

- I enjoy conversing with the chatbot.
- The chatbot is friendly.
- I've learned interesting things from the chatbot.

Curiosity

- When I have a language question in mind while conversing with the chatbot, I cannot rest without knowing the answer.
- I wonder how well I can speak English after conversing with the chatbot.
- I wonder how it would feel to speak English as fluently as the chatbot does.

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